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ASSIGNMENT -8

Session based Shopping Cart using Java Servlets and HttpSession

AIM

To develop a **session-based shopping cart web application using Java Servlets and HttpSession** that allows users to add products, store cart items in a session, view cart contents, remove items, and calculate subtotal and grand total dynamically.

CODE

Product.html

```
<!DOCTYPE html>
<html>
<head>
  <title>Products</title>
  <style>
    body {
      font-family: Arial, sans-serif;
      margin: 20px;
    }
    h2 {
      color: #333;
    }
    form {
      border: 1px solid #ccc;
      padding: 15px;
      margin-bottom: 20px;
      width: 300px;
    }
    input[type="number"] {
      width: 50px;
    }
    a {
      color: #007BFF;
    }
  </style>
</head>
<body>
  <h2>Product List</h2>
  <form action="Add_to_cart_servlet" method="post">
```

```

        <h3>Laptop - 50000</h3>
        <input type="hidden" name="name" value="Laptop">
        <input type="hidden" name="price" value="50000">
        Quantity:
        <input type="number" name="quantity" value="1" min="1">
        <br><br>
        <input type="submit" value="Add to Cart">
    </form>
    <br>

    <form action="Add_to_cart_servlet" method="post">
        <h3>Mobile - 20000</h3>
        <input type="hidden" name="name" value="Mobile">
        <input type="hidden" name="price" value="20000">
        Quantity:
        <input type="number" name="quantity" value="1" min="1">
        <br><br>
        <input type="submit" value="Add to Cart">
    </form>
    <br>

    <form action="Add_to_cart_servlet" method="post">
        <h3>Headphones - 3000</h3>
        <input type="hidden" name="name" value="Headphones">
        <input type="hidden" name="price" value="3000">
        Quantity:
        <input type="number" name="quantity" value="1" min="1">
        <br><br>
        <input type="submit" value="Add to Cart">
    </form>

    <br><br>
    <a href="View_cart_servlet" style="text-decoration: none;">View Cart</a>

</body>
</html>

```

CartItem.java

```

import java.io.Serializable;

public class CartItem implements Serializable {
    private String name;
    private double price;
    private int quantity;

    public CartItem(String name, double price, int quantity) {

```

```

        this.name = name;
        this.price = price;
        this.quantity = quantity;
    }
    public String getName() {
        return name;
    }
    public void setName(String name) {
        this.name = name;
    }
    public double getPrice() {
        return price;
    }
    public void setPrice(double price) {
        this.price = price;
    }
    public int getQuantity() {
        return quantity;
    }
    public void setQuantity(int quantity) {
        this.quantity = quantity;
    }
}

```

Add_to_cart Servlet.java

```

import java.io.IOException;
import java.util.ArrayList;
import java.util.List;
import jakarta.servlet.ServletException;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;

public class Add_to_cart_servlet extends HttpServlet {

    protected void doPost(HttpServletRequest request, HttpServletResponse
response)throws ServletException, IOException {

        String name = request.getParameter("name");
        double price = Double.parseDouble(request.getParameter("price"));
        int quantity = Integer.parseInt(request.getParameter("quantity"));
    }
}

```

```

        HttpSession session = request.getSession();
        List<CartItem> cart = (List<CartItem>) session.getAttribute("cart");
        if (cart == null) {
            cart = new ArrayList<CartItem>();
        }

        boolean found = false;
        for (CartItem item : cart) {
            if (item.getName().equals(name)) {
                item.setQuantity(item.getQuantity() + quantity);
                found = true;
                break;
            }
        }

        if (!found) {
            CartItem item = new CartItem(name, price, quantity);
            cart.add(item);
        }
        session.setAttribute("cart", cart);
        response.sendRedirect("products.html");
    }
}

```

View cart Servlet.java

```

import java.io.IOException;
import java.io.PrintWriter;
import java.util.List;

import jakarta.servlet.ServletException;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;

public class View_cart_servlet extends HttpServlet {

    protected void doGet(HttpServletRequest request, HttpServletResponse
response)throws ServletException, IOException {

        response.setContentType("text/html");
        PrintWriter out = response.getWriter();
        HttpSession session = request.getSession();
        List<CartItem> cart = (List<CartItem>) session.getAttribute("cart");
        double grandTotal = 0;
        out.println("<!DOCTYPE html>");
    }
}

```

```

        out.println("<html>");
        out.println("<head>");
        out.println("<title>Shopping Cart</title>");
        out.println("<style>");
        out.println("table { width:50%; border-collapse:collapse; }");
        out.println("th, td { border:1px solid black; padding:8px; text-align:left;
});

        out.println("th { background-color:#f2f2f2; }");
        out.println("a { color:#007BFF; text-decoration:none; }");
        out.println("</style>");

        out.println("</head>");
        out.println("<body>");
        out.println("<h2>Your Cart</h2>");
        if (cart == null || cart.size() == 0) {
            out.println("<p>Cart is empty</p>");
        } else {

            out.println("<table>");
            out.println("<tr>");
            out.println("<th>Name</th>");
            out.println("<th>Price</th>");
            out.println("<th>Quantity</th>");
            out.println("<th>Subtotal</th>");
            out.println("<th>Action</th>");
            out.println("</tr>");

            for (CartItem item : cart) {

                double subtotal = item.getPrice() * item.getQuantity();
                grandTotal += subtotal;
                out.println("<tr>");
                out.println("<td>" + item.getName() + "</td>");
                out.println("<td>" + item.getPrice() + "</td>");
                out.println("<td>" + item.getQuantity() + "</td>");
                out.println("<td>" + subtotal + "</td>");
                out.println("<td>");
                out.println("<a href='Remove_from_cart_servlet?name=" + item.getName()
+ "'>Remove</a>");
                out.println("</td>");
                out.println("</tr>");
            }

            out.println("</table>");
            out.println("<h3>Grand Total: " + grandTotal + "</h3>");
        }

        out.println("<br>");

```

```

        out.println("<a href='products.html' style=\"text-decoration: none;\">Back to  
Products</a>");

        out.println("</body>");
        out.println("</html>");
    }
}

```

Remove_from_cart_servlet.java

```

import java.io.IOException;
import java.util.Iterator;
import java.util.List;

import jakarta.servlet.ServletException;
import jakarta.servlet.http.HttpServlet;
import jakarta.servlet.http.HttpServletRequest;
import jakarta.servlet.http.HttpServletResponse;
import jakarta.servlet.http.HttpSession;

public class Remove_from_cart_servlet extends HttpServlet {

    protected void doGet(HttpServletRequest request, HttpServletResponse
response)throws ServletException, IOException {

        String name = request.getParameter("name");
        HttpSession session = request.getSession();
        List<CartItem> cart = (List<CartItem>) session.getAttribute("cart");
        if (cart != null) {
            Iterator<CartItem> iterator = cart.iterator();
            while (iterator.hasNext()) {
                CartItem item = iterator.next();
                if (item.getName().equals(name)) {
                    iterator.remove();
                    break;
                }
            }
        }

        session.setAttribute("cart", cart);
        response.sendRedirect("View_cart_servlet");
    }
}

```

Web.xml

```
<web-app>

    <servlet>
        <servlet-name>Add_to_cart_servlet</servlet-name>
        <servlet-class>Add_to_cart_servlet</servlet-class>
    </servlet>

    <servlet-mapping>
        <servlet-name>Add_to_cart_servlet</servlet-name>
        <url-pattern>/Add_to_cart_servlet</url-pattern>
    </servlet-mapping>

    <servlet>
        <servlet-name>View_cart_servlet</servlet-name>
        <servlet-class>View_cart_servlet</servlet-class>
    </servlet>

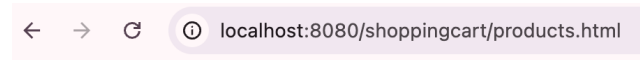
    <servlet-mapping>
        <servlet-name>View_cart_servlet</servlet-name>
        <url-pattern>/View_cart_servlet</url-pattern>
    </servlet-mapping>

    <servlet>
        <servlet-name>Remove_from_cart_servlet</servlet-name>
        <servlet-class>Remove_from_cart_servlet</servlet-class>
    </servlet>

    <servlet-mapping>
        <servlet-name>Remove_from_cart_servlet</servlet-name>
        <url-pattern>/Remove_from_cart_servlet</url-pattern>
    </servlet-mapping>

</web-app>
```

OUTPUT



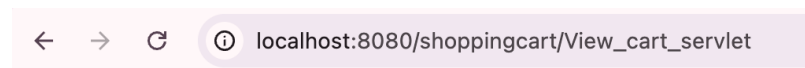
Product List

Laptop - 50000
Quantity:

Mobile - 20000
Quantity:

Headphones - 3000
Quantity:

[View Cart](#)



Your Cart

Name	Price	Quantity	Subtotal	Action
Laptop	50000.0	1	50000.0	Remove
Mobile	20000.0	1	20000.0	Remove
Headphones	3000.0	3	9000.0	Remove

Grand Total: 79000.0

[Back to Products](#)

← → ↻ ⓘ localhost:8080/shoppingcart/View_cart_servlet

Your Cart

Name	Price	Quantity	Subtotal	Action
Mobile	20000.0	1	20000.0	Remove
Headphones	3000.0	3	9000.0	Remove

Grand Total: 29000.0

[Back to Products](#)

← → ↻ ⓘ localhost:8080/shoppingcart/View_cart_servlet

Your Cart

Cart is empty

[Back to Products](#)

LEARNING OUTCOME

- **Understand how** Java Servlets handle HTTP requests and responses **in a web application.**
- **Learn how to use** HttpSession to store and manage user session data **such as cart items.**
- **Implement a** model class (POJO) with serialization **to represent cart items containing name, price, and quantity.**
- **Develop the ability to** process form data and dynamically generate HTML responses using servlets.
- **Gain experience in** building a simple shopping cart system with item addition, removal, quantity handling, and total price calculation.